

Literature Status for Q8 of arXiv:0901.3029

FA/Banach run packet

June 19, 2026

Status

Result type: literature already answered.

Source problem: Assadi–Haghshenas, *Some Classical and Recent Results Concerning Renorming Theory*, arXiv:0901.3029, Q8 in the final “Some interesting problems” list, PDF page 14.

Supporting answer: Hajek–Russo–Somaglia–Todorćević, *An Asplund space with norming Markushevich basis that is not weakly compactly generated*, arXiv:2007.14201, Problem 1.1 and Theorem A, PDF page 2.

Source Question

The source survey asks:

(Q8) (Godefroy) Assume an Asplund space X has a Markushevich basis $\{x_i, f_i\}_{i \in I}$ with $\text{span}\{f_i\}_{i \in I}$ norming X^* . Is X WCG?

This is the norming Markushevich-basis versus weakly compactly generated problem attributed to Godefroy.

Supporting Answer

The later paper arXiv:2007.14201 restates the same problem as Problem 1.1: if X is an Asplund space with a norming Markushevich basis, must X be weakly compactly generated? It says that the problem was recorded in several sources, including GMZ Problem 112, and that the main result is a negative answer.

Theorem A of arXiv:2007.14201 states that there exists an Asplund space X with a 1-norming M-basis such that X is not WCG. This supplies a direct counterexample to Q8 of arXiv:0901.3029.

Scope and Verification

The identification is direct: the supporting authors explicitly know they are answering Godefroy’s norming-M-basis/WCG problem, and their theorem negates the exact conclusion asked in Q8. No part of Q8 remains open under this reading. This packet does not address any of the other questions in arXiv:0901.3029.

Local checks used the TeX sources and PDFs for arXiv:0901.3029 and arXiv:2007.14201. Cheap run indexes were searched for arXiv:0901.3029, arXiv:2007.14201, Godefroy, norming Markushevich basis, and WCG. A previous attempt note had already found the answer paper; this packet promotes the status only because arXiv:0901.3029 provides a separate extracted open-problem source.

References

References

- [1] A. Assadi and H. Haghshenas, *Some Classical and Recent Results Concerning Renorming Theory*, arXiv:0901.3029. <https://arxiv.org/abs/0901.3029>
- [2] P. Hajek, T. Russo, J. Somaglia, and S. Todorcevic, *An Asplund space with norming Markushevich basis that is not weakly compactly generated*, arXiv:2007.14201. <https://arxiv.org/abs/2007.14201>